4D spacetime

Notation and Setup:

Starting Metric for this Case: g_{00}=A[t, r], g_{11}=-a[t]^2 B[r], g_{22}=-r^2 a[t]^2 C[r], g_{33}=-r^2 a[t]^2 Sin[θ]^2 C[r], a(t) not constant

Corresponding Line Element

ds^2 = A[t, r] dt^2 - a[t]^2 B[r] dr^2 - r^2 a[t]^2 C[r] dθ^2 - r^2 a[t]^2 Sin[θ]^2 C[r] dϕ^2

Output Display Simplification - Parameters are sometimes suppressed

Prime (') means derivative wrt r.

Dot (.) means derivative wrt t.

Examples: **A = A(t,r)**, **A'=∂_rA(t,r)** and **$\dot{A}$=∂_tA(t,r)**

Lambda vacuum equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$. The $g_{\mu\nu}$ matrix used is a diagonal matrix. Since off-diagonal $R_{\mu\nu}$ equations rarely appear, almost all the equations are expressed by dividing out $g_{\mu\nu}$.

FLRW Equations: An exact solution to the Λ vacuum field equations that has (1) free parameter a, (2) C = 1, and (3) parameter $\mathcal{A}$ such that $\mathcal{A}$(t) = c^2 is the Friedmann Continuity Equation (which generates the a(t) formula).

Free-gauge FLRW (FG-FLRW) Equations: An exact solution to the Λ vacuum field equations that has (1) free parameters a and C, and (2) parameter $\mathcal{A}$ such that $\mathcal{A}$(t) = c^2 is the Friedmann Continuity Equation (which generates the a(t) formula).

a: Free-gauge FLRW Solution

We generalize the FLRW solution by letting $\mathcal{A}$, B, and C be free parameters.

More specifically, we begin with the following assumptions:

A(t,r) is allowed to be a function of t and r,

B(r) is strictly a function of r,

C(r) is a free parameter,

a(t) is not a constant.

With this starting point, the non-zero Lambda vacuum field equations are:

$$R_{00}: \quad \frac{3\,\ddot{a}}{a\,A} - \frac{3\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{a^2\,A\,r\,B} + \frac{(A')^2}{4\,a^2\,A^2\,B} - \frac{A'\,C'}{2\,a^2\,A\,C\,B} + \frac{A'\,B'}{4\,a^2\,A\,B^2} - \frac{A''}{2\,a^2\,A\,B} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{a}\,A'}{a\,A} = 0$$

$$R_{11}: \quad \frac{2\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} + \frac{(A')^2}{4\,a^2\,A^2\,B} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{(C')^2}{2\,a^2\,C^2\,B} + \frac{B'}{a^2\,r\,B^2} + \frac{A'\,B'}{4\,a^2\,A\,B^2} + \frac{C'\,B'}{2\,a^2\,C\,B^2} - \frac{A''}{2\,a^2\,A\,B} - \frac{C''}{a^2\,C\,B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2\,C} - \frac{1}{a^2\,r^2\,B} + \frac{2\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{2\,a^2\,A\,r\,B} - \frac{2\,C'}{a^2\,r\,C\,B} - \frac{A'\,C'}{4\,a^2\,A\,C\,B} + \frac{B'}{2\,a^2\,r\,B^2} + \frac{C'\,B'}{4\,a^2\,C\,B^2} - \frac{C''}{2\,a^2\,C\,B} = \Lambda$$

$$R_{33}: \quad \frac{1}{a^2\,r^2\,C} - \frac{1}{a^2\,r^2\,B} + \frac{2\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{2\,a^2\,A\,r\,B} - \frac{2\,C'}{a^2\,r\,C\,B} - \frac{A'\,C'}{4\,a^2\,A\,C\,B} + \frac{B'}{2\,a^2\,r\,B^2} + \frac{C'\,B'}{4\,a^2\,C\,B^2} - \frac{C''}{2\,a^2\,C\,B} = \Lambda$$

R_{33} is redundant, the same as R_{22}.

Since $a(t) \neq$ constant, from R_{01} we get that

$A(t,r) = \mathcal{A}(t)$, a new unknown, strictly a function of t.

R_{01} is eliminated, and the remaining equations become

$$R_{00}: \quad \frac{3\,\ddot{a}}{a\,\mathcal{A}} - \frac{3\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} = \Lambda$$

$$R_{11}: \quad \frac{2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{\ddot{a}}{a\,\mathcal{A}} - \frac{\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{(C')^2}{2\,a^2\,C^2\,B} + \frac{B'}{a^2\,r\,B^2} + \frac{C'\,B'}{2\,a^2\,C\,B^2} - \frac{C''}{a^2\,C\,B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2\,C} - \frac{1}{a^2\,r^2\,B} + \frac{2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{\ddot{a}}{a\,\mathcal{A}} - \frac{\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{B'}{2\,a^2\,r\,B^2} + \frac{C'\,B'}{4\,a^2\,C\,B^2} - \frac{C''}{2\,a^2\,C\,B} = \Lambda$$

Solving R_{00} for $\mathcal{A}$ yields

$$\mathcal{A}(t) = \frac{3\,\dot{a}(t)^2}{\Lambda\,a(t)^2 - 3\,k} \quad \text{where } k \in \mathbb{R} \text{ is a constant of integration.}$$

This leaves 2 equations in the 2 unknowns, B and C:

$$R_{11}: \quad -\frac{2\,k}{a^2} + \Lambda - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{(C')^2}{2\,a^2\,C^2\,B} + \frac{B'}{a^2\,r\,B^2} + \frac{C'\,B'}{2\,a^2\,C\,B^2} - \frac{C''}{a^2\,C\,B} = \Lambda$$

$$R_{22}: \quad -\frac{2\,k}{a^2} + \frac{1}{a^2\,r^2\,C} - \frac{1}{a^2\,r^2\,B} + \Lambda - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{B'}{2\,a^2\,r\,B^2} + \frac{C'\,B'}{4\,a^2\,C\,B^2} - \frac{C''}{2\,a^2\,C\,B} = \Lambda$$

Λ cancels out of both equations, and we can then solve for B algebraically:

$$R_{11}-2R_{22}:\quad \frac{2k}{a^2}+\frac{2}{a^2 B r^2}-\frac{2}{a^2 r^2 C}+\frac{2C'}{a^2 B r C}+\frac{(C')^2}{2a^2 B C^2}-\Lambda = -\Lambda$$

$$B(r) = -\frac{(r C'(r)+2C(r))^2}{4C(r)\left(k r^2 C(r)-1\right)}$$

k is the curvature constant. k can be any real number, positive, zero, or negative.

Plugging this in makes the 2 remaining equations into identities:

R_{11}: $\Lambda = \Lambda$

R_{22}: $\Lambda = \Lambda$

All of the equations are satisfied and so we have an exact solution to Λ field equations:

$$\mathcal{A}[t] = \frac{3\dot{a}[t]^2}{-3k+\Lambda a[t]^2},\quad \text{where k is a free constant,}$$

$$B(r) = -\frac{(2C[r]+rC'[r])^2}{4C[r]\left(-1+k r^2 C[r]\right)},$$

$C(r)$ is a free parameter,

and line element

$$ds^2 = \frac{3\dot{a}(t)^2}{\Lambda a(t)^2-3k}dt^2+\frac{a(t)^2(rC'(r)+2C(r))^2}{4C(r)\left(k r^2 C(r)-1\right)}dr^2-r^2 a(t)^2 C(r)\,d\theta^2$$
$$-r^2 a(t)^2 \sin^2(\theta) C(r)\,d\phi^2$$

In Section c we will show that the FLRW solution is a special case of this solution.

I call this solution (having free parameter C) the Free-gauge FLRW (FG-FLRW) solution.

In these notebooks the name FLRW is used only for 4D. Similar solutions in 5D-7D are referred to as Generalized Robertson-Walker (GRW), and solutions having free parameter C are referred to as Free-gauge Generalized Robertson-Walker (FG-GRW).

Observations:

- A was allowed to be a function of both t and r. It is a result of the field equations, not ansatz, that A(t,r) = $\mathcal{A}$(t) is strictly a function of t.
- It is a somewhat surprising find that there is a generalization of the FLRW solution that has C as a free parameter.

b: Solve $\mathcal{A}(t) = c^2$ to find a(t)

Solving $\frac{3\,\dot{a}^2}{a^2\,\Lambda - 3\,k} = c^2$ for $a(t)$ where $\dot{a} = \frac{da}{dt}$.

Recall the definitions of $\cosh^{-1}$ and $\sinh^{-1}$:

$$\cosh^{-1}(\omega) = \int \frac{d\omega}{\sqrt{\omega^2 - 1}} \quad \text{and} \quad \sinh^{-1}(\omega) = \int \frac{d\omega}{\sqrt{\omega^2 + 1}}$$

We begin with,

$$\mathcal{R}(t) = \frac{3\,\dot{a}^2}{-3\,k + a^2\,\Lambda} = c^2$$

$$\Rightarrow \quad \dot{a}^2 = c^2 \left(-k + \frac{a^2\,\Lambda}{3}\right)$$

$$\Rightarrow \quad \frac{da}{dt} = \dot{a} = c\,\sqrt{-k + \frac{a^2\,\Lambda}{3}}$$

If $k \neq 0$, this can be expressed

$$\frac{da}{dt} = \sqrt{k}\,c\,\left(\sqrt{\frac{\Lambda}{3\,k}\,a^2 - 1}\right) = \begin{cases} \sqrt{|k|}\,c\,\sqrt{\frac{\Lambda}{3|k|}\,a^2 - 1} & \text{if } k > 0 \\ \sqrt{|k|}\,c\,\sqrt{\frac{\Lambda}{3|k|}\,a^2 + 1} & \text{if } k < 0 \end{cases}$$

If we define

$$\omega = \sqrt{\frac{\Lambda}{3\,|k|}}\,a,$$

then

$$\frac{da}{dt} = \begin{cases} \sqrt{|k|}\,c\,\sqrt{\omega^2 - 1} & \text{if } k > 0 \\ \sqrt{|k|}\,c\,\sqrt{\omega^2 + 1} & \text{if } k < 0 \end{cases}$$

Differentiating both sides of the ω equations yields

$$\sqrt{\Lambda}\,da = \sqrt{3\,|k|}\,d\omega$$

Multiplying both sides of the $\frac{da}{dt}$ equation by $\sqrt{\Lambda}\,dt$ and rearranging, we get

$$\sqrt{\Lambda}\,\sqrt{|k|}\,c\,dt = \begin{cases} \frac{\sqrt{\Lambda}\,da}{\sqrt{\omega^2-1}} = \sqrt{\frac{3|k|}{\sqrt{\omega^2-1}}}\,d\omega & \text{if } k > 0 \\ \frac{\sqrt{\Lambda}\,da}{\sqrt{\omega^2+1}} = \sqrt{\frac{3|k|}{\sqrt{\omega^2+1}}}\,d\omega & \text{if } k < 0 \end{cases}$$

Since $k \neq 0$, $\sqrt{|k|}$ can be divided out of both sides:

$$\sqrt{\Lambda}\ c\ dt = \begin{cases} \frac{\sqrt{3}}{\sqrt{\omega^2-1}}\,d\omega & \text{if } k > 0 \\ \frac{\sqrt{3}}{\sqrt{\omega^2+1}}\,d\omega & \text{if } k < 0 \end{cases}$$

$$\Rightarrow \quad \sqrt{\Lambda}\ c\ t = \sqrt{\Lambda}\ c \int dt = \begin{cases} \sqrt{3}\int \frac{d\omega}{\sqrt{\omega^2-1}} = \sqrt{3}\cosh^{-1}[\omega] = \sqrt{3}\cosh^{-1}[\sqrt{\frac{\Lambda}{3\,|k|}}\,a] & \text{if } k > 0 \\ \sqrt{3}\int \frac{d\omega}{\sqrt{\omega^2+1}} = \sqrt{3}\sinh^{-1}[\omega] = \sqrt{3}\sinh^{-1}[\sqrt{\frac{\Lambda}{3\,|k|}}\,a] & \text{if } k < 0 \end{cases}$$

where the constant of integration, an offset, has been set to zero.

$$\Rightarrow \quad \sqrt{\frac{\Lambda}{3}}\,c\,t = \begin{cases} \cosh^{-1}[\sqrt{\frac{\Lambda}{3\,|k|}}\,a] & \text{if } k > 0 \\ \sinh^{-1}[\sqrt{\frac{\Lambda}{3\,|k|}}\,a] & \text{if } k < 0 \end{cases}$$

Eliminating the inverse functions, we get

$$\sqrt{\frac{\Lambda}{3\ |k|}}\,a = \begin{cases} \cosh[\sqrt{\frac{\Lambda}{3}}\,ct] & \text{if } k > 0 \\ \sinh[\sqrt{\frac{\Lambda}{3}}\,ct] & \text{if } k < 0 \end{cases}$$

or

(1) $a(t) = \sqrt{\frac{3\,|k|}{\Lambda}}\cosh[\sqrt{\frac{\Lambda}{3}}\,ct] \quad \text{if } k > 0,\ \Lambda \neq 0$

(2) $a(t) = \sqrt{\frac{3\,|k|}{\Lambda}}\sinh[\sqrt{\frac{\Lambda}{3}}\,ct] \quad \text{if } k < 0$

Formula (2) is valid when $\Lambda = 0$:

$$\frac{1}{\sqrt{\Lambda}}\,\text{Sinh}[\sqrt{\frac{\Lambda}{3}}\,ct] = \lim_{\Lambda\to 0}\frac{\text{Sinh}\left[\sqrt{\frac{\Lambda}{3}}\,c\,t\right]}{\sqrt{\Lambda}} = \frac{c\,t}{\sqrt{3}} \quad \text{due to L'Hospital's Rule}$$

which implies

(3) $a(t) = c\,t\,\sqrt{|k|}$

In finding solutions (1-3) we have assumed $k \neq 0$. When $k = 0$ (flat space), $\mathcal{R}(t) = c^2$ becomes:

$$\mathcal{R}(t) = \frac{3\,\dot{a}^2}{a^2\,\Lambda} = c^2$$

The result is

(4) $a(t) = e^{\pm\frac{c\,t\,\sqrt{\Lambda}}{\sqrt{3}}}\,\alpha$ where α is a constant of integration

Combining the solutions, we have:

$$\textbf{(5)}\quad a(t) = \begin{cases} \sqrt{\frac{3|k|}{\Lambda}}\cosh\left[\sqrt{\frac{\Lambda}{3}}\,ct\right] & \text{if } k > 0 \;\&\; \Lambda \neq 0 \\ \sqrt{\frac{3|k|}{\Lambda}}\sinh\left[\sqrt{\frac{\Lambda}{3}}\,ct\right] & \text{if } k < 0 \;\&\; \Lambda \neq 0 \\ c\,t\,\sqrt{|k|} & \text{if } k < 0 \;\&\; \Lambda = 0 \\ \alpha\, e^{\pm\frac{c\,t\,\sqrt{\Lambda}}{\sqrt{3}}} & \text{if } k = 0 \;\&\; \Lambda \geq 0, \text{ where } \alpha > 0 \text{ is a free constant} \end{cases}$$

There is no solution when $k = 0$ and $\Lambda < 0$.

c: FLRW Solution

Plugging $C(r)=1$ into the Case a solution produces the following solution:

$$\mathcal{A}[t] = \frac{3\,\dot{a}[t]^2}{-3\,k + \Lambda\,a[t]^2} \quad \text{where k is a constant,}$$

$$B(r) = \frac{1}{1 - k\,r^2} \quad \text{where } kr^2 < 1,$$

$$C(r) = 1,$$

and line element

$$ds^2 = \frac{3\,\dot{a}[t]^2}{-3\,k + \Lambda\,a[t]^2}\,dt^2 - \frac{a[t]^2}{1 - k\,r^2}\,dr^2 - r^2\,a[t]^2\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,d\phi^2$$

We will show in Section d that $\mathcal{A}(t)=c^2$ is the Friedmann continuity equation.

- The solution of the continuity equation is $a(t)$.

Thus, **our preferred way to display the FLRW equations is** the above solution with a slight change to the display of $\mathcal{A}$:

$$\mathcal{A}[t] = \frac{3\,\dot{a}[t]^2}{-3\,k + \Lambda\,a[t]^2} = c^2 \quad \text{where k is a constant,}$$

$$B(r) = \frac{1}{1 - k\,r^2} \quad \text{where } kr^2 < 1,$$

$$C(r) = 1,$$

Consider the standard way to display the FLRW equations: $\mathcal{A}(t) = c^2$ (without the $\mathcal{A}(t)$ formula) and either include $a(t)$, as below, or else implicitly assume the Friedmann equations:

$$a(t) = \begin{cases} \sqrt{\frac{3|k|}{\Lambda}}\cosh\left[\sqrt{\frac{\Lambda}{3}}\,ct\right] & \text{if } k > 0 \;\&\; \Lambda \neq 0 \\ \sqrt{\frac{3|k|}{\Lambda}}\sinh\left[\sqrt{\frac{\Lambda}{3}}\,ct\right] & \text{if } k < 0 \;\&\; \Lambda \neq 0 \\ c\,t\,\sqrt{|k|} & \text{if } k < 0 \;\&\; \Lambda = 0 \\ \alpha\, e^{\pm\frac{c\,t\,\sqrt{\Lambda}}{\sqrt{3}}} & \text{if } k = 0 \;\&\; \Lambda \geq 0, \text{ where } \alpha > 0 \text{ is a free constant} \end{cases}$$

$$\mathcal{A}(t) = c^2,$$

$$B(r) = \frac{a[t]^2}{1 - k\,r^2},$$

$$C(r) = a[t]^2,$$

and line element

$$ds^2 = c^2\,dt^2 - \frac{a[t]^2}{1 - k\,r^2}\,dr^2 - r^2\,a[t]^2\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,d\phi^2$$

Observation:

- The $B(r) = \frac{1}{1 - k\,r^2}$ FLRW metric term requires a reduced domain, $kr^2 < 1$, in order that $B(r) > 0$.
- This is known as **Reduced Circumference Polar Coordinates (RCPC)**.
- RCPC restricts the domain of r (for $k > 0$, $0 \le r < \frac{1}{\sqrt{k}}$).
- k itself remains a free constant.
- Analogous domain restrictions on r appear in higher dimensions.

d: The Vacuum Friedmann Equations

Plugging $\mathcal{A}=c^2$, B, and C (but not a) from the FLRW solution into the field equations yields:

R_{00}: $\frac{3\,\ddot{a}}{a\,c^2} = \Lambda$

R_{11}: $\frac{2\,k}{a^2} + \frac{2\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{22}: $\frac{2\,k}{a^2} + \frac{2\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{22} is redundant, the same as R_{11}.

The Friedmann continuity and acceleration equations, respectively, are the following pair of equations generated by algebraic manipulation of R_{00} and R_{11}:

$-\frac{1}{6}R_{00} + \frac{1}{2}R_{11}$: $\frac{k}{a^2} + \frac{\dot{a}^2}{a^2\,c^2} = \frac{\Lambda}{3}$

$\frac{1}{4}R_{00} + \frac{1}{4}R_{11}$: $\frac{k}{2\,a^2} + \frac{\dot{a}^2}{2\,a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \frac{\Lambda}{2}$

We can easily see that $\mathcal{A}(t) = c^2$ is simply a rearrangement of the continuity equation. Thus, the 4-branch a(t) formula is the solution to the Friedmann equations.